Keyword while Cond^n :

$\boxed{\text{statement}}$

statement

True

False

None

if = 4

Keyword

Module

math

import math

144 → math.sqrt(144) ←

~~Keyword~~

random

~~datetime~~

matplotlib

numpy

panda

(1, 100)

import random

y = f(x)

f( )

→ help ('modules')

~~Task~~

Random number guess game

1. random number generate → 76
2. input from user (x) → 77
3. x > random number
   ↳ you number is greater
4. x < random number
   ↳ your number is smaller
5. (x == random number) → "You won"

Loop

```
import random

jackpot=random.randint(1,100)
guess=int(input("Guess your number: "))
while True:
    if guess<jackpot:
        print('Wrong!Guess higher')
    else:
        print("Wrong! Guess Lower")
    guess=int(input("Guess you number:"))
```

76

→ 4

F

$\boxed{\text{guess != jackpot}}$

→ print

jackpot	guess
$\boxed{76}$	4
	80
	$\boxed{76}$

Wrong! Guess higher

Wrong! Guess Lower

Binary Search

Jackpot = 76

1 50 75 100

87

81 ~~87~~

75 81

75 78

76, 77

$\frac{100+50}{2}$

$\frac{175}{2}$

75
87
162
$\frac{162}{2}$

6

2cr - search

1cr

NID

1 2 3 . . . . . . 16cr

8cr

1 — 1

4cr

2

3 — 2cr

4 — 1cr

for loop

for variable in [ ] :

x

0 to (5-1)

iteration

```
for i in range(5):
    print(i) →
```

i=0

0

i=1

i=2

range(1, 100, +2)

list [1, 3, 5, 7, 9, ... 99]

```
for i in range(1,100,+2):
    Print(i)
```

1
3
5

$$\frac{1}{1!} + \frac{2}{2!} + \frac{3}{3!} + \cdots \frac{n}{n!}$$

user → n → i/p

$$1 \leftarrow \frac{1}{1!} + \frac{2}{2!} + \frac{3}{3!} =$$

```
result = 0
fact = 1
for i in range (1,n+1):
    fact= fact * i
    result = result + i/fact
```

1	2	3

result	fact
~~0~~	~~1~~
~~1~~	~~1~~
2	2

iteration-01

$i=1$

$fact = 1*1 = 1$

$result = 0 + 1/1 = 1$

iteration-02

$i=2$

$fact = 1*2 = 2$

$result = 1 + 2/2 = 2$

$$\frac{1^2}{1!} + \frac{2^2}{2!} + \frac{3^2}{3!} + \frac{4^2}{4!} + \cdots + \frac{n^2}{n!}$$

Nested Loop

	1	2	3
1	(1,1)	(1,2)	(1,3)
2	(2,1)	(2,2)	(2,3)
3	(3,1)	(3,2)	(3,3)

1 1
1 2
1 3
1 4
2 1

row → 1 → n
n → $n*n$ → n^2

```
for row in range(1,4):
    for col in range(1,4):
        print(row,col)
```

```
for row in range(1,n+1):
    for col in range(1,n+1):
        print(row,col)
```

row | 1 | 2 | 3 |

iteration-01

row = 1

col | 1 | 2 | 3 |

itr-01
col 1
iter-02
col → 2
iter-03
col → 3

iteration-02

row = 2

col | 1 | 2 | 3 |

1, 1
1 2
1 3
2 1
2 2
2 3
3 1
3 2
3 3

Pattern-01

n=4

```
*
* *
* * *
* * * *
```

$\boxed{n=3}$

```
*
* *
* * *
```

n=5

```
*
* *
* * *
* * * *
* * * * *
```

Pattern-02

row = 3
col = 4

```
* * * *
* * * *
* * * *
```

range(row) → 0 to (row-1)

range(1,row) → 1 to (row-1)

range(1, row+1) → 1 to row

range(1, row+1, +1) → 1 to row

how many numbers / row

row-1

row

row

r = 4
c = 2

```
* *
* *
* *
* *
```

row = 4   col = 2

```
row=int(input("Enter row: "))
col=int(input("Enter col: "))
for r in range(row):
    for c in range(col):
        print("*",end="")
```

r = $\boxed{0,1,2,3}$

iter-01

r=0

→ c $\boxed{0 \mid 1}$

print()

iteration-02

r=1

→ c $\boxed{0 \mid 1}$

```
* * *
* *
```

1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4

n=4

n=5

1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5

n=4

i 1 2 3 4

i=1

j= 1 2 3 4

1 2 3 4
1 2 3 4
1 2 3 4

```
n=int(input("Enter n: "))

for i in range(1,n+1):
    for j in range(1,n+1):
        print(j,end="")
    Print()
```

i=2

j 1 2 3 4

0,1

ASCII

A → 65
B → 66
C → 67
.
.
Z → .

Print(65)

Print(65)

char(65)

Pattern-05

n=4

1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4

Pattern-06

n=4

A A A A
B B B B
C C C C
D D D D

Pattern-07

a a a a
b b b b
c c c c
d d d d

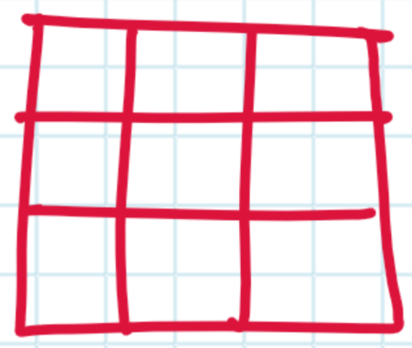
Pattern-08

n=4

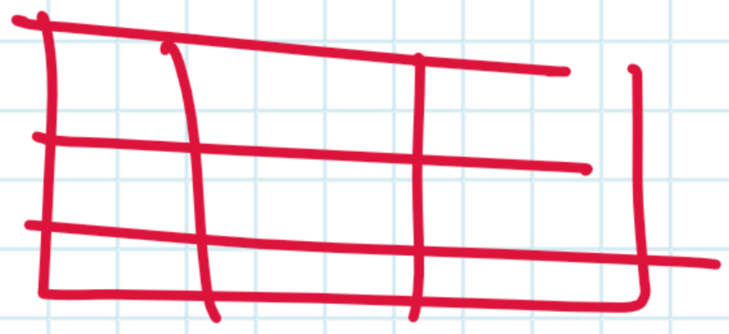
1
1 2
1 2 3
1 2 3 4

*
* *
* * *
* * * *

```
for i in range(n ):
    for j in range(n):
        print(i)
```



```
for row in range(row):
    for c in range(col):
        Print( )
```



```
for i in range(n):
    for j in range(i):
```

Pattern-09

A
A B
A B C

Pattern-10

n=4

1
A B
1 2 3
A B C D

Pattern-11

n=4

* * * *
* * *
* *
*

Pattern-12

n=4

1 2 3 4
1 2 3
1 2
1

13

a b c d
a b c
a b
a

14

row, col
5 7

* * * * * * *
* # # # # # *
* # # # # # *
* # # # # # *
* * * * * * *

15

$n=5$

```
- - * - -
- - * - -
* * * * *
- - * - -
- - * - -
```

16

```
* - - - *
- * - * -
- - * - -
- * - * -
* - - - *
```

17

$n=4$

```
1
2 3
4 5 6
7 8 9 10
```

18

```
1
1 3
1 3 5
1 3 5 7
```